

Total (3+3 questions)

Common data: Use the following common data if not mentioned specifically in the question

For $0.25\mu\text{m}$ Technology node -- $L_{\min}=250\text{ nm}$, $\text{pico}=10^{-12}$, $\text{femto}=10^{-15}$

Take $V_{DD}=2.5\text{ V}$, $V_{T0N}=0.4\text{V}$, $V_{T0P}=-0.4\text{V}$, $R_{n\text{mos}}=13\text{ k}\Omega/(\text{W/L})$, $R_{p\text{mos}}=31\text{ k}\Omega/(\text{W/L})$, $\lambda_n=\lambda_p=0.1\text{ V}^{-1}$, $\mu_n C_{ox}=90\text{ }\mu\text{A/V}^2$, $\mu_p C_{ox}=30\text{ }\mu\text{A/V}^2$

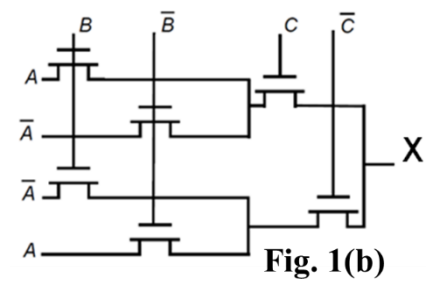
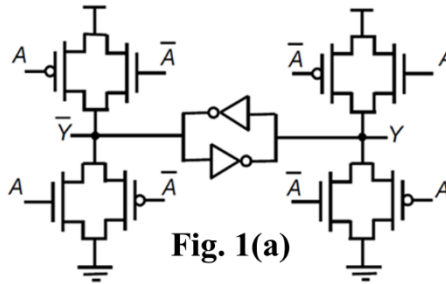
$g(\text{inv})=1$, $p(\text{inv})=1$ for standard cmos inverter,

- Use long channel approximation and lambda-based design rules unless specially mentioned.
- Use square law current equation unless specifically mentioned. Neglect channel length modulation in current calculation, if not mentioned specifically. The body of all PMOS is tied to V_{DD} while all NMOS is tied to ground unless specially mentioned.

Answer the sub-parts of a question at one place and in the order in which they appear.

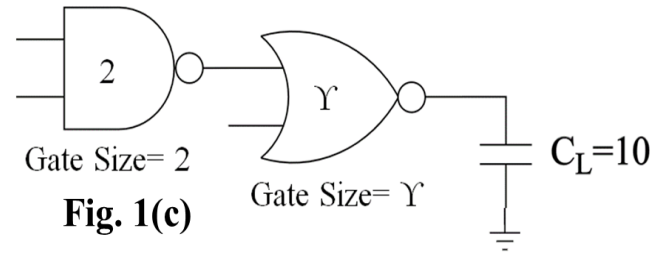
SECTION - A (55 marks total)

Q1. (a). Determine the Boolean Functions Y and $\bar{Y}$ implemented by the circuit shown in Fig. 1(a) and Boolean Functions X implemented in Fig. 1(b).



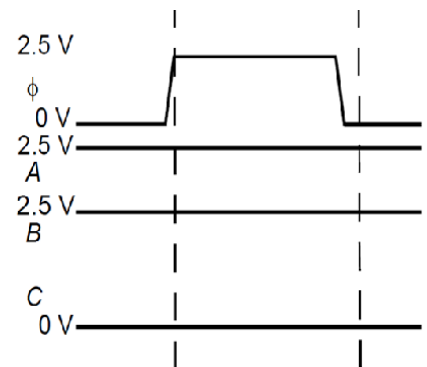
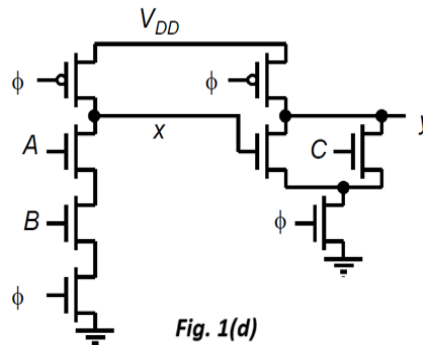
Q1. (b) Given that minimum sized standard CMOS inverter has $W_N=1$ and $W_P=2$, (take this gate has gate size=1). Take $\mu_p = 2\mu_n$

Determine the value of Y for minimum path delay -of the 2-stage circuit shown in Fig. 1(c).



Also, determine minimum path delay in delay units.

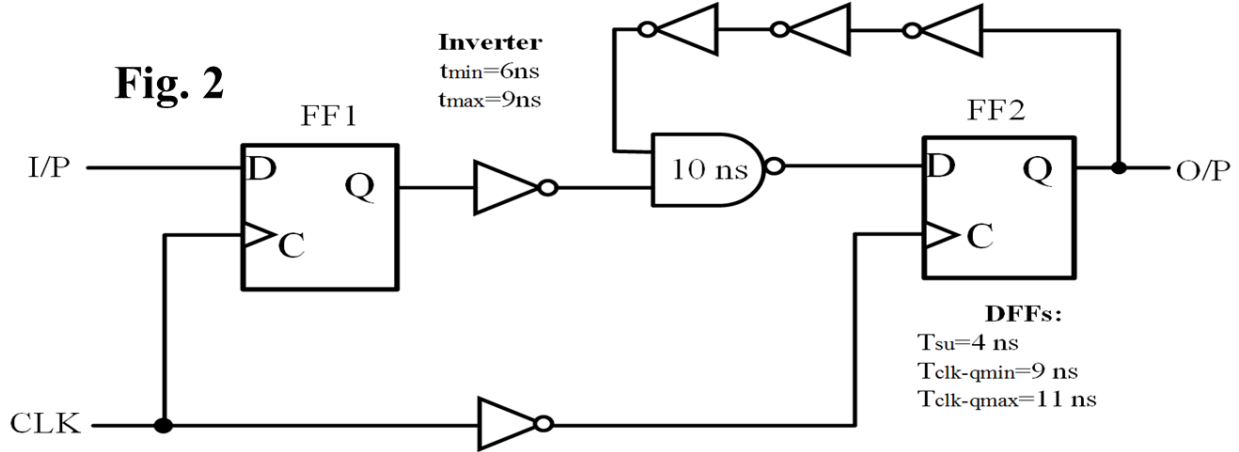
Q1. (c) Consider the circuits shown in Figure 1 (d) and Figure 1 (e). For the given input combinations, plot node voltages V_X and V_Y .



[19 marks]

P.T.O.

Q2. Consider the circuit shown in **Fig. 2**, where the **flip-flop delays** and **gate delays** are annotated. Perform a detailed timing analysis of the circuit and answer the following:



Determine the maximum frequency of operation for the circuit in MHz. Show all steps and calculations.

- Find the hold time requirement** for the circuit. Provide a detailed explanation and calculations to support your answer.
- Now, Assume a **clock skew of 9 ns** is introduced between **Flip-Flop 1 (FF1)** and **Flip-Flop 2 (FF2)**. Analyze whether a hold time violation occurs under this condition. Clearly, calculate the hold time constraint and explain your reasoning.

[18 marks]

Q3. Consider the design of a CMOS compound (OAI22) gate in a **22 nm** technology node to implement the Boolean function appended below:

$$F = \bar{A} \cdot \bar{B} + \bar{C} \cdot \bar{D}$$

Given that the complimentary versions of inputs are not available, $(W/L)_n=20$, and $(W/L)_p=3(W/L)_n$.

- Sketch its transistor-level schematic.
- Sketch the corresponding stick diagram and properly label the source, polysilicon, & drain region of each transistor.
- Estimate the cell area from the stick diagram.
- Calculate the parasitic junction capacitance at the output node. (Here, please ignore the overlap capacitance in the channel region. The area junction capacitance is $C_j = 3 \times 10^{-4} \text{ pF}/\mu\text{m}^2$, and the side wall junction capacitance is $C_{jsw} = 2 \times 10^{-4} \text{ pF}/\mu\text{m}$). Consider the length of the source and drain regions as per the λ -based design rules tabulated below.
- Draw the cross-sectional diagram of the transistors connected in the n-diffusion region.

[18 marks]

λ -based design rules:

Active contact size	2λ
Minimum active contact to metal edge spacing	λ
Minimum active contact to poly edge spacing	3λ
Minimum active contact to active edge spacing	λ

-----Section A ends-----

Write the section name clearly on top of each sheet.
Answer the sub-parts of a question at one place and in the order in which they appear.

SECTION - B (Total 50 marks)

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μ m Technology node -- $L_{min}=250$ nm,

pico (p) = 10^{-12} , femto (f) = 10^{-15}

Neglect body effect

Take $V_{DD} = 2.5$ V, $V_{T0N}=0.4$ V, $V_{T0P} = -0.4$ V, $\lambda_n = \lambda_p = 0.1$ V $^{-1}$, $\mu_n C_{ox} = 90$ μ A/V 2 , $\mu_p C_{ox} = 30$ μ A/V 2

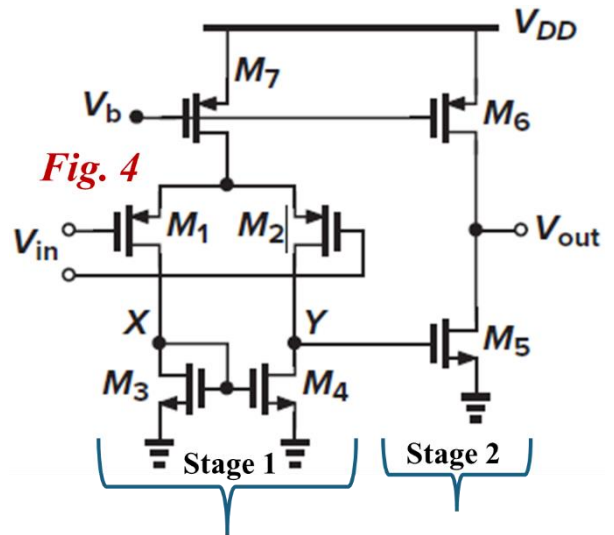
Q4. For the 2-stage op amp shown in **Fig. 4**, assume a power budget of 6 mW, a required output swing of 2.5 V, and $L_{eff} = 0.5$ μ m for all devices.

Given, $V_{DD} = 3$ V

for nMOS: $\lambda = 0.1$ V $^{-1}$, $V_{Tn} = 0.7$ V, $\mu_n C_{ox} = 134$ μ A/V 2 .

For pMOS: $\lambda = 0.1$ V $^{-1}$, $V_{Tp} = -0.7$ V, $\mu_p C_{ox} = 40$ μ A/V 2 .

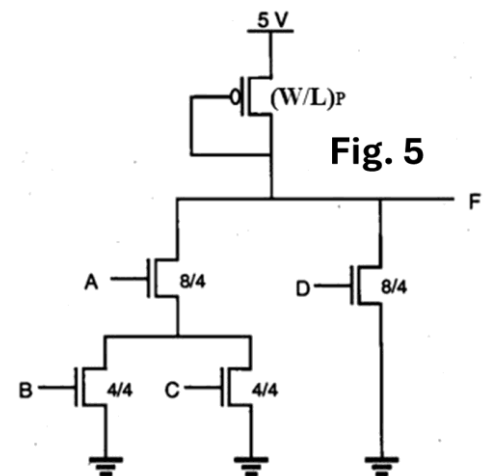
- Allocating a current of 1 mA to the output stage and equal overdrive voltages to M_5 and M_6 , determine $(W/L)_5$ and $(W/L)_6$.
- Calculate the small-signal gain of the output stage (V_{out}/V_y).
- Determine the aspect ratio (W/L) of M_3 and M_4 such that $V_{GS3} = V_{GS5}$.
- Calculate the aspect ratios of M_1 and M_2 such that the overall voltage gain of the op amp is equal to 500.



[16 marks]

Q5. A) Consider the logic circuit shown in **Fig. 5**, with threshold voltage V_{To} (**driver**) = 1 V, V_{To} (**load**) = -3 V, and body effect coefficient (γ) = 0.

- Determine the logic function F.
- Calculate $(W/L)_P$ such that minimum output voltage (V_{OL}) does not exceed 0.4 V.



P.T.O.

B) Consider a CMOS inverter with the following parameters:

Given: $V_{DD} = 1.2 \text{ V}$, the channel length of both transistors is $L_n = L_p = 60 \text{ nm}$.

nMOS $V_{T0,n} = 0.48 \text{ V}$, $\mu_n C_{ox} = 102 \mu\text{A/V}^2$, $E_{c,n} L_n = 0.4 \text{ V}$, $(W/L)_n = 10$

pMOS $V_{T0,p} = -0.46 \text{ V}$, $\mu_p C_{ox} = 51.6 \mu\text{A/V}^2$, $E_{c,p} L_p = 1.8 \text{ V}$, $(W/L)_p = 19$

i) . Determine the (W_n / W_p) ratio so that the switching (inversion) threshold voltage of the circuit is $V_{th} = 0.5 \text{ V}$.

ii) . The CMOS fabrication process used to manufacture this inverter allows a variation of the $V_{T0,n}$ value by $\pm 15\%$ around its nominal value, and a variation of the $V_{T0,p}$ value by $\pm 20\%$ around its nominal value.

Assuming that all other parameters (such as μ_n , μ_p , C_{ox} , W_n , W_p) always retain their nominal values, find the upper and lower limits of the **switching threshold voltage (V_{th})** of this circuit.

[17 marks]

Q6. Consider shown in **Fig. 6**. Take $V_{ov} = 0.2 \text{ V}$. Assume capacitive load at node V1 is **14 pF**, and capacitive load at node Vout is **2 pF**

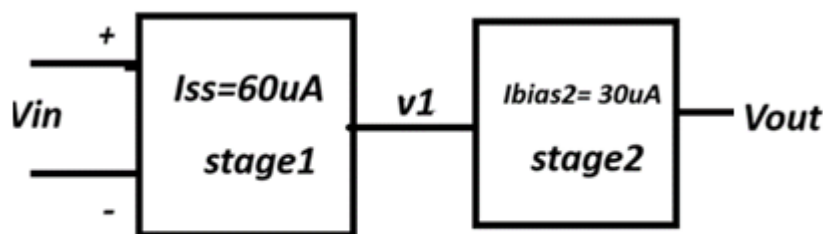


Fig. 6

a) Sketch and label the schematic of **Fig. 6** for the following cases—

- Case-1**---- stage1 is current mirror load differential amplifier, and stage2 is pmos input cascode amplifier
- Case-2**---- stage1 is current mirror load differential amplifier with cascode load, and stage2 is pmos input common source amplifier

Intuitively---

b) Determine low frequency small signal voltage gain $A_v = v_{out} / V_{in}$ for both **case-1**, and **case-2**

c) Determine the pole frequencies at node v1, and node Vout for both **case-1**, and **case-2**

d) Sketch and label Bode magnitude plot for both **case-1**, and **case-2**

[17 marks]

------(End section B)-----